Sheet: VeraModule

VERA FPGA

File: vera-fpga.sch

Sheet: BusDecoder

BUS DECODER

File: busdecoder.sch

Sheet: Vera FPGA flash

VERA SPI FLASH
SD CARD INTERFACE

File: vera-fpga-flash.sch

Sheet: A2SlotInterface

APPLE II SLOT INTERFACE

File: a2slotInterface.sch

Sheet: PowerSupply

POWER SUPPLY

File: powersupply.sch

Sheet: VGA Analog

VGA ANALOG

File: vga-analog.sch

FIDUCIAL TOP

FID3 Fiducial

FID4 Fiducial

FIDUCIAL BOTTOM

FID2 Fiducial

FID1 Fiducial

LOGO1
VERA X16 LOGO

LOGO2
RETROCAMPUS

LOGO3
KiCAD DESIGN

LOGO5
ESP32_BOBOARD

LBL1

H1 MountingHole

H2 MountingHole

H3 MountingHole

H4 MountingHole

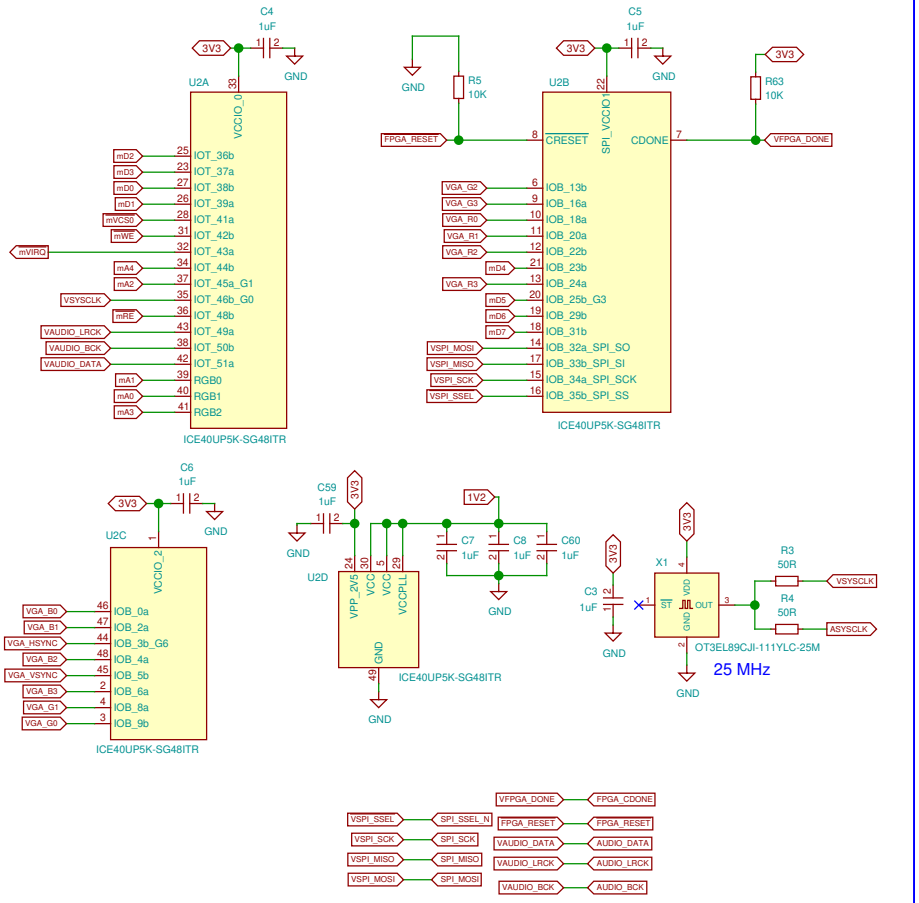
Gianluca Renzi
RetroBit Lab

Sheet: /
File: VERA-MODULE-RBL.sch

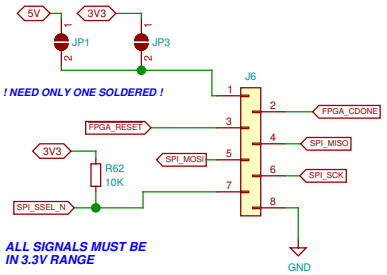
Title: VERA FPGA Audio & Video Board

Size: A4	Date: 2026-11-14	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261112-20:34Z		Id: 1/7

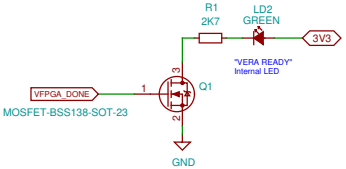
FPGA VERA LOGIC VIDEO & AUDIO CARD



iceprog programmer USB FTDI / SPI

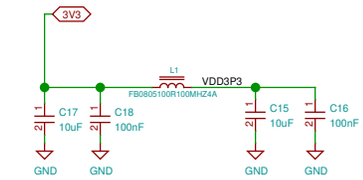


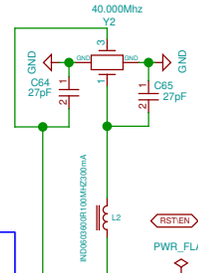
VERA FPGA PROGRAMMED OK

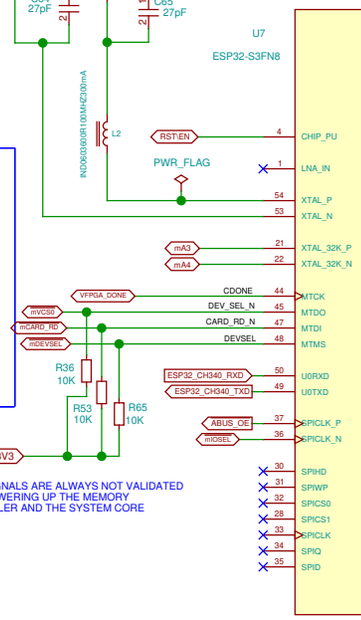


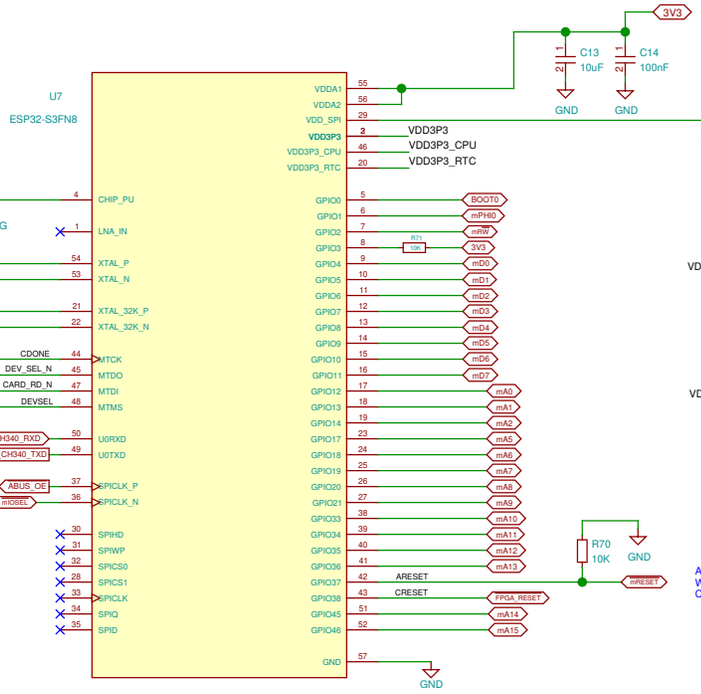
Gianluca Renzi		
RetroBit Lab		
Sheet: /VeraModule/		
File: vera-fpga.sch		
Title: VeraModule		
Size: A4	Date: 2026-11-14	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261112-20:34Z		Id: 2/7

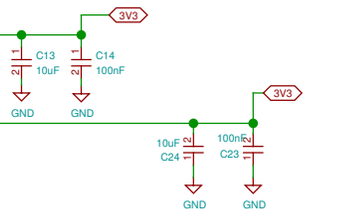
SLOT NUMBER: decoded by the Apple (/DEVSEL, /IOSEL)





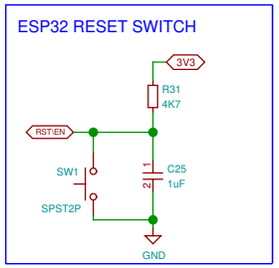


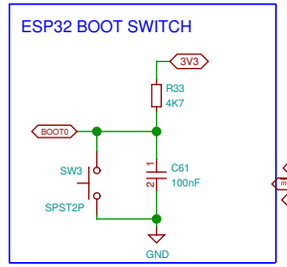




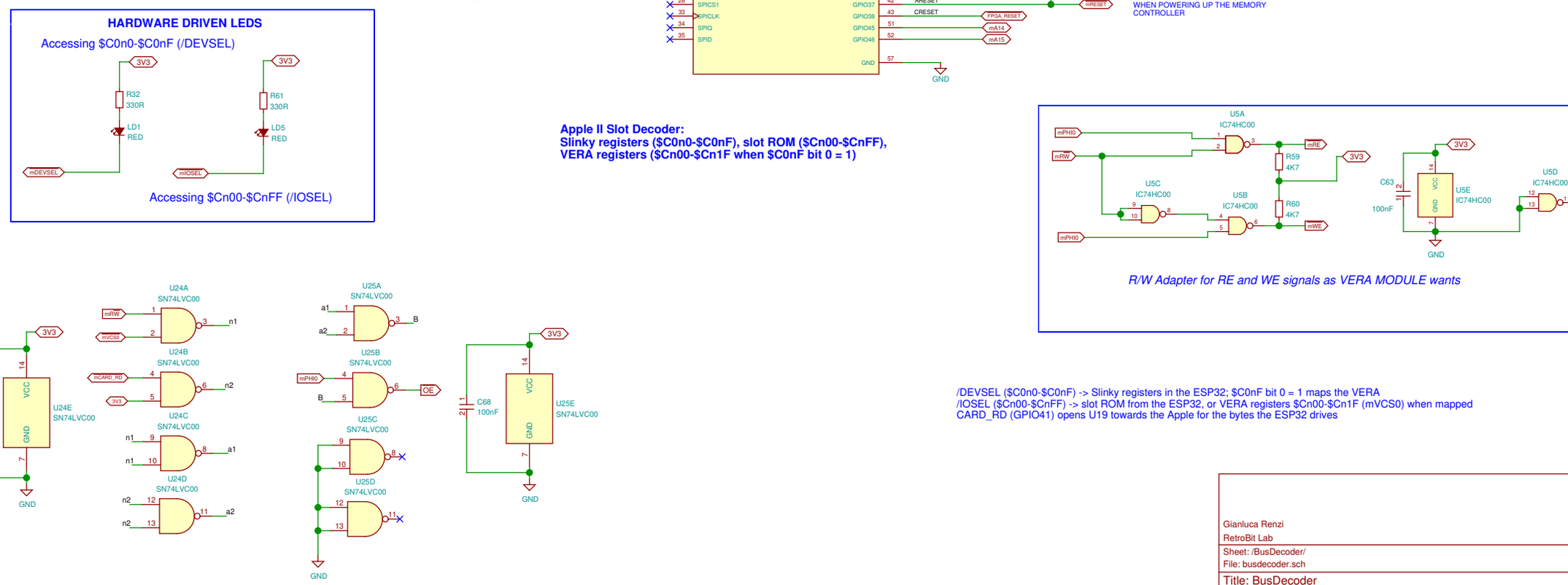


APPLE RESET IS ALWAYS ASSERTED WHEN POWERING UP THE MEMORY CONTROLLER





APPLE SIGNALS ARE ALWAYS NOT VALIDATED WHEN POWERING UP THE MEMORY CONTROLLER AND THE SYSTEM CORE



Apple II Slot Decoder:
Slinky registers ($C0n0-$C0nF), slot ROM ($Cn00-$CnFF),
VERA registers ($Cn00-$Cn1F when $C0nF bit 0 = 1)

R/W Adapter for RE and WE signals as VERA MODULE wants

/DEVSEL ($C0n0-$C0nF) -> Slinky registers in the ESP32; $C0nF bit 0 = 1 maps the VERA
/IOSEL ($Cn00-$CnFF) -> slot ROM from the ESP32, or VERA registers $Cn00-$Cn1F (mVCS0) when mapped
CARD_RD (GPIO41) opens U19 towards the Apple for the bytes the ESP32 drives

Gianluca Renzi
RetroBit Lab

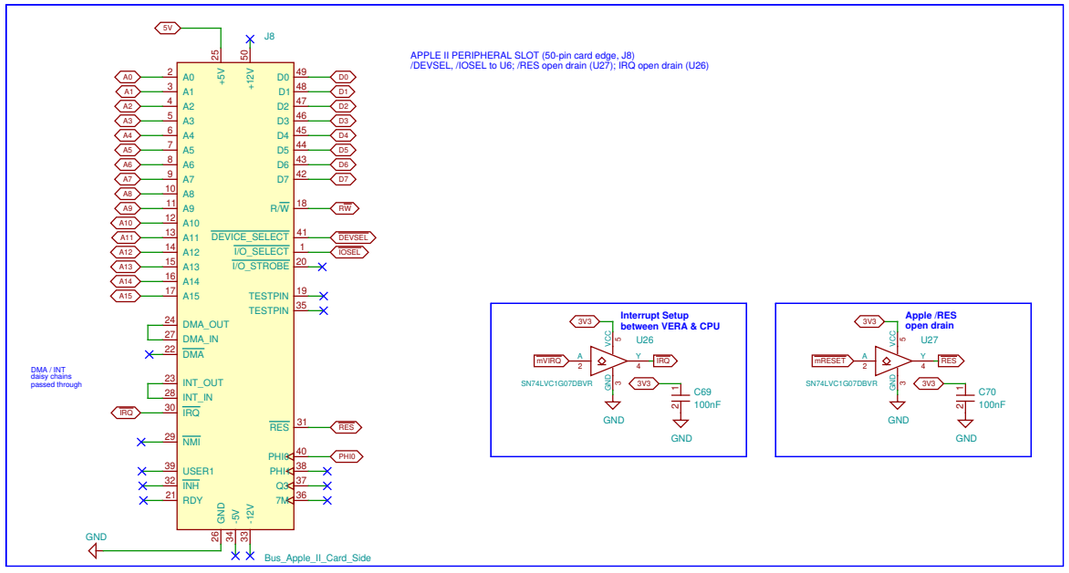
Sheet: /BusDecoder/
File: busdecoder.sch

Title: BusDecoder

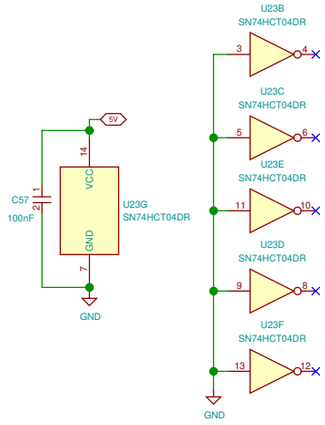
Size: A3	Date: 2026-11-14	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261112-20:34Z		Id: 3/7

APPLE II PERIPHERAL SLOT INTERFACE

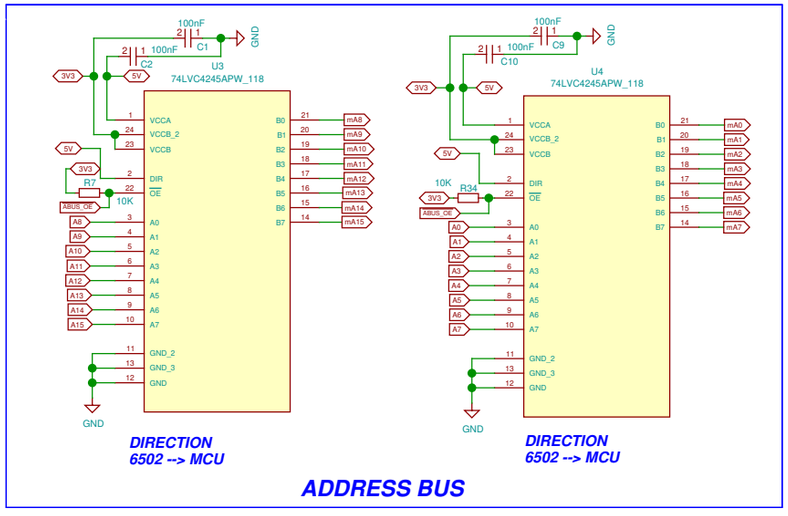
All 8-Bit signals must be shifted from 5V to 3.3V and vice versa

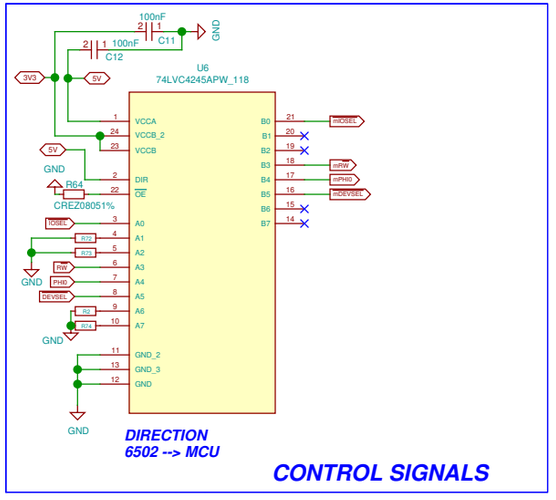


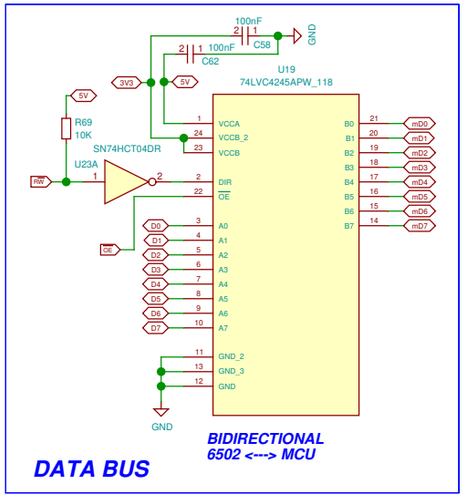




BUS LOGIC LEVEL SHIFTERS







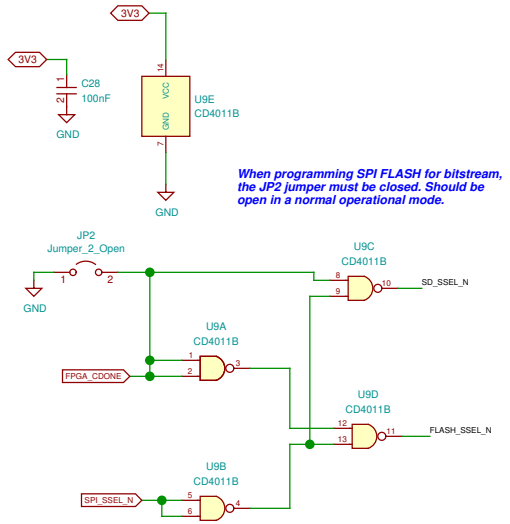
Gianluca Renzi
RetroBit Lab

Sheet: /A2SlotInterface/
File: a2slotInterface.sch

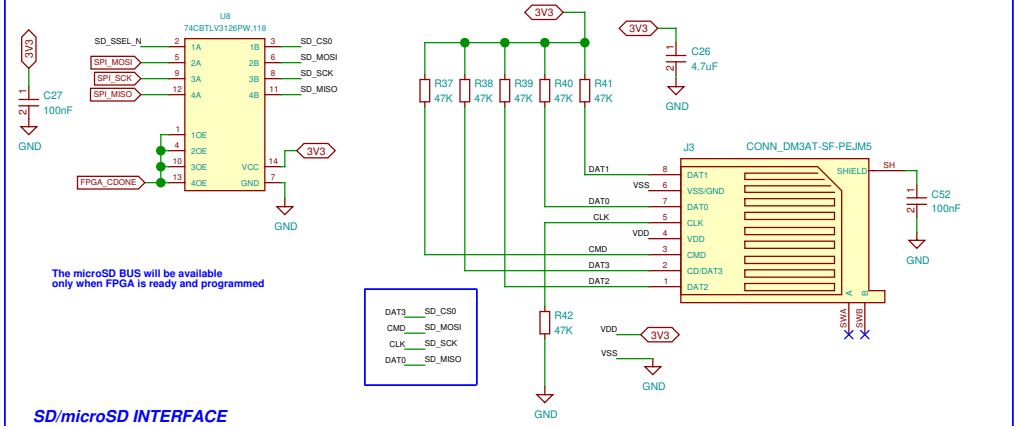
Title: A2SlotInterface

Size: A3	Date: 2026-11-14	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261112-20:34Z		Id: 4/7

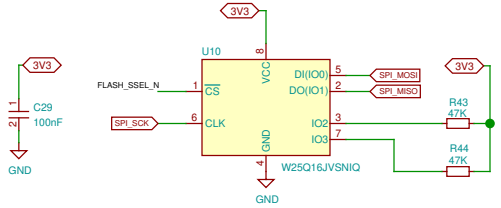
FPGA/SSD Flash Glue Logic



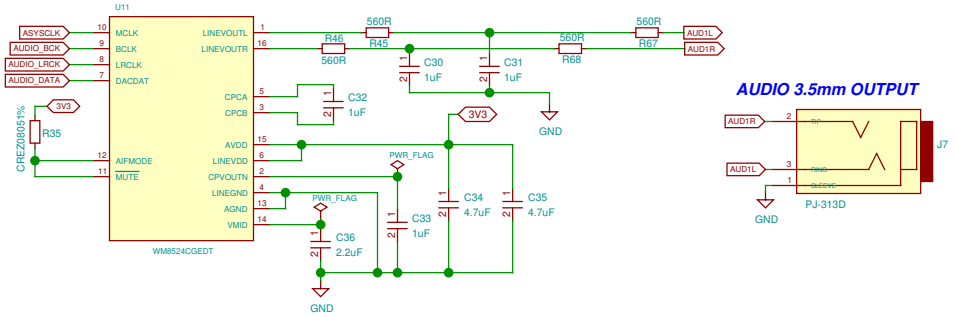
SD/microSD INTERFACE



SPI 16MB FLASH



IC DAC/AUDIO 24BIT 192K 16TSSOP



AUDIO 3.5mm OUTPUT

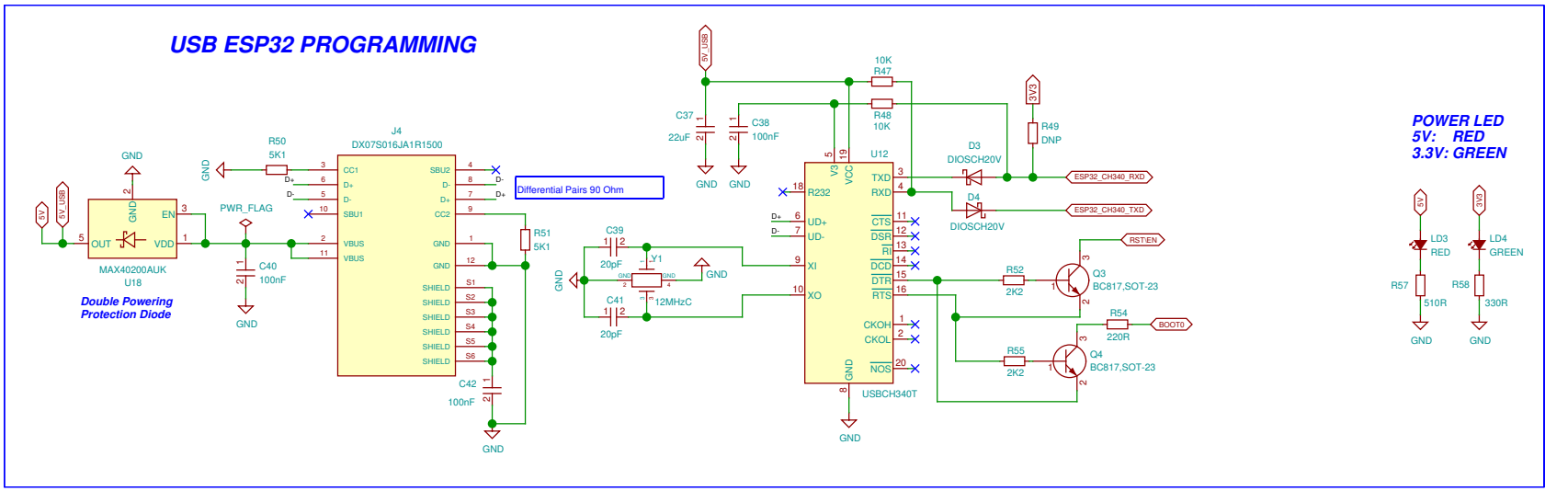
AUDIO SECTION

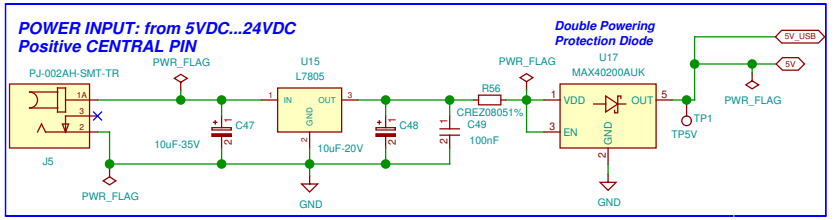
Gianluca Renzi
RetroBit Lab

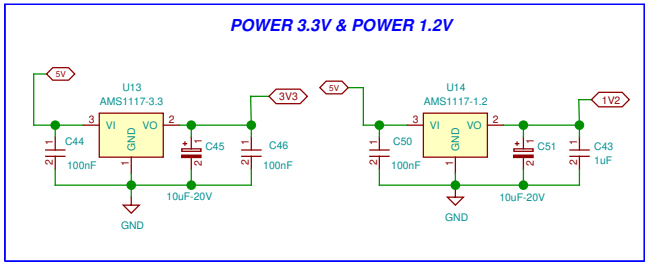
Sheet: /Vera FPGA flash/
File: vera-fpga-flash.sch

Title: Vera FPGA flash

Size: A4	Date: 2026-11-14	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261112-20:34Z		Id: 5/7

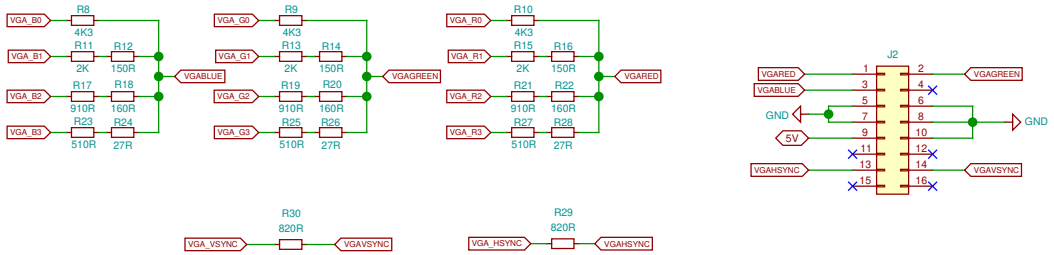




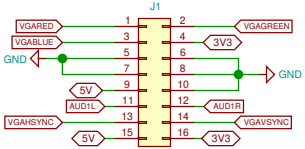


Gianluca Renzi	
RetroBit Lab	
Sheet: /PowerSupply/	
File: powersupply.sch	
Title: PowerSupply	
Size: A4	Date: 2026-11-14
Plotted by eeshow 3a954ac+ 20261112-20:34Z	Rev: 1.0
	Id: 6/7

ANALOG VGA SIGNALS



EXTERNAL VIDEO CONNECTOR



EXTERNAL DISPLAY WITH AUDIO

Gianluca Renzi		
RetroBit Lab		
Sheet: /VGA Analog/		
File: vga-analog.sch		
Title: VGA Analog		
Size: A4	Date: 2026-11-14	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261112-20:34Z		Id: 7/7